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JEE MAIN · QUESTION PAPER

JEE Main 2026 (Online) 24th January Evening Shift

JEE Main Question Paper with Answer Key

24 Jan 2026	75	English
HELD ON	QUESTIONS	LANGUAGE

INSIDE THIS PDF

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Answer key	at the end

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Chemistry 25 questions

Q1. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement I : There are several conformers for n -butane. Out of those conformers,



Statement II : As the dihedral angle increases, torsional strain decreases from (X) to (Y).

In the light of the above statements, choose the correct answer from the options given below

(A) Statement I is false but Statement II is true

(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false

(D) Both Statement I and Statement II are true

Q2. MCQ (Single Correct Answer)

Open solution

In the Group analysis of cations, Ba^{2+} & Ca^{2+} are precipitated respectively as

(A) sulphide & sulphide

(B) hydroxide & carbonate

(C) chromate & sulphide

(D) carbonate & carbonate

Q3. MCQ (Single Correct Answer)

Open solution

" X " is an oxoanion of the lightest element of group 7 (in the periodic table). The metal is in +6 oxidation state in " X ". The color of the potassium salt of X is

(A) purple

(B) orange

(C) yellow

(D) green

Q4. MCQ (Single Correct Answer)

Open solution

At 298 K , the mole percentage of $\mathrm{N}_2(\mathrm{~g})$ in air is 80%. Water is in equilibrium with air at a pressure of 10 atm . What is the mole fraction of $\mathrm{N}_2(\mathrm{~g})$ in water at 298 K ? (K_{H} for N_2 is $6.5 \times 10^7 \mathrm{~mmHg}$)

(A) 9.35×10^{-5}

(B) 9.35×10^5

(C) 1.23×10^{-7}

(D) 1.17×10^{-4}

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Q5. MCQ (Single Correct Answer)

Open solution

The heat of atomisation of methane and ethane are ' x ' $\mathrm{kJmol^{-1}}$ and ' y ' $\mathrm{kJmol^{-1}}$ respectively. The longest wavelength (λ) of light capable of breaking the $\mathrm{C-C}$ bond can be expressed in SI unit as :

(A) $\frac{\mathrm{N_A hc}}{250(y-6x)}$

(B) $\mathrm{N_A hc}\left(y-\frac{6x}{4}\right)^{-1}$

(C) $\frac{\mathrm{hc}}{1000}\left(\frac{y-6x}{4}\right)^{-1}$

(D) $\frac{\mathrm{N_A hc}}{250(4y-6x)}$

Q6. MCQ (Single Correct Answer)

Open solution

Pair of species among the following having same bond order as well as paramagnetic character will be-

(A) $\mathrm{O_2^-, N_2^+}$

(B) $\mathrm{{O_2}^+, {N_2}^{2-}}$

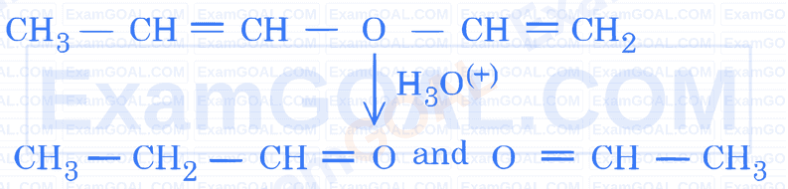
(C) $\mathrm{O_2^-, N_2^-}$

(D) $\mathrm{O_2^+, N_2^-}$

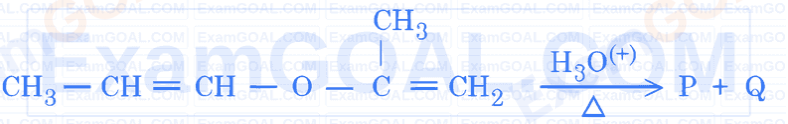
Q7. MCQ (Single Correct Answer)

Open solution

The unsaturated ether on acidic hydrolysis produces carbonyl compounds as shown below :-



Based on this, predict the solution/reagent that will help to distinguish "P" and "Q" obtained in the following reaction :-



(A) Saturated $\mathrm{NaHSO_3}$ solution

(B) Fehling solution

(C) Lucas reagent

(D) 2, 4 - DNP reagent

Q8. MCQ (Single Correct Answer)

Open solution

Consider the following gaseous equilibrium in a closed container of volume ' V ' at $\mathrm{T(K)}$.

$\mathrm{P_2(g) + Q_2(g) \rightleftharpoons 2PQ(g)}$

2 moles each of $\mathrm{P_2(g)}$, $\mathrm{Q_2(g)}$ and $\mathrm{PQ(g)}$ are present at equilibrium. Now one mole each of ' $\mathrm{P_2}$ ' and ' $\mathrm{Q_2}$ ' are added to the equilibrium keeping the temperature at $\mathrm{T(K)}$. The number of moles of $\mathrm{P_2, Q_2}$ and PQ at the new equilibrium, respectively, are

(A) $2.56, 1.62, 2.24$

(B) $2.67, 2.67, 2.67$

(C) $1.21, 2.24, 1.56$

(D) $1.66, 1.66, 1.66$

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Q9. MCQ (Single Correct Answer)

The correct order of $\mathrm{C}, \mathrm{N}, \mathrm{O}$ and F in terms of second ionisation potential is

(A) $\mathrm{C} < \mathrm{N} < \mathrm{F} < \mathrm{O}$

(B) $\mathrm{C} < \mathrm{F} < \mathrm{N} < \mathrm{O}$

(C) $\mathrm{F} < \mathrm{N} < \mathrm{C} < \mathrm{O}$

(D) $\mathrm{C} < \mathrm{O} < \mathrm{N} < \mathrm{F}$

Open solution

Q10. MCQ (Single Correct Answer)

Given below are two statements :

Statement I : Cross aldol condensation between two different aldehydes will always produce four different products.

Statement II : When semicarbazide reacts with a mixture of benzaldehyde and acetophenone under optimum pH , it forms a condensation product with acetophenone only.

In the light of the above statements, choose the correct answer from the options given below

(A) Statement I is true but Statement II is false

(B) Both Statement I and Statement II are false

(C) Both Statement I and Statement II are true

(D) Statement I is false but Statement II is true

Open solution

Q11. MCQ (Single Correct Answer)

Choose the **INCORRECT** statement

(A) Carbon exhibits negative oxidation states along with +4 and +2 .

(B) Carbon cannot exceed its covalency more than four.

(C) Among the isotopes of carbon, $^{13}\mathrm{C}$ is a radioactive isotope.

(D) CO_2 is the most acidic oxide among the dioxides of group of 14 elements.

Open solution

Q12. MCQ (Single Correct Answer)

The wavelength of spectral line obtained in the spectrum of Li^{2+} ion, when the transition takes place between two levels whose sum is 4 and difference is 2 , is

(A) $1.14 \times 10^{-6}\ \mathrm{cm}$

(B) $2.28 \times 10^{-6}\ \mathrm{cm}$

(C) $1.14 \times 10^{-7}\ \mathrm{cm}$

(D) $2.28 \times 10^{-7}\ \mathrm{cm}$

Open solution

Q13. MCQ (Single Correct Answer)

Two liquids A and B form an ideal solution at temperature T K . At T K , the vapour pressures of pure A and B are 55 and $15\mathrm{kNm}^{-2}$ respectively. What is the mole fraction of A in solution of A and B in equilibrium with a vapour in which the mole fraction of A is 0.8?

(A) 0.5217

(B) 0.480

(C) 0.340

(D) 0.663

Open solution

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Q14. MCQ (Single Correct Answer)

From the following, how many compounds contain at least one secondary alcohol?



Open solution

Choose the correct answer from the options given below :

(A) Three (B) Two (C) Four (D) Five

Q15. MCQ (Single Correct Answer)

Given below are two statements :

Statement I : The dipole moment of R-CN is greater than R-NC and R-NC can undergo hydrolysis under acidic medium to produce $R-\overset{\overset{O}{\|}}{C}-OH$.

Statement II : R-CN hydrolyses under acidic medium to produce a compound which on treatment with $SOCl_2$, followed by the addition of NH_3 gives another compound (x). This compound (x) on treatment with $NaOCl/NaOH$ gives a product, that on treatment with $CHCl_3/KOH/\Delta$ produces R-NC

In the light of the above statements, choose the correct answer from the options given below

(A) Both Statement I and Statement II are false

(B) Statement I is false but Statement II is true

(C) Statement I is true but Statement II is false

(D) Both Statement I and Statement II are true

Open solution

Q16. MCQ (Single Correct Answer)

The number of possible tripeptides formed involving alanine (ala), glycine (gly) and valine (val), where no amino acid has been used more than once is:

(A) 8 (B) 3 (C) 4 (D) 6

Open solution

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Q17. MCQ (Single Correct Answer)

Open solution

One mole of $Cl_2(g)$ was passed into 2 L of cold 2 M KOH solution. After the reaction, the concentrations of Cl^-, ClO^- and OH^- are respectively (assume volume remains constant)

(A) $0.5M, 0.5M, 0.5M$

(B) $1M, 1M, 1M$

(C) $0.5M, 0.5M, 1M$

(D) $0.75M, 0.75M, 1M$

Q18. MCQ (Single Correct Answer)

Open solution

A student has planned to prepare acetanilide from aniline using acetic anhydride.

The student has started from 9.3 g of aniline. However, the student has managed to obtain 11 g of dry acetanilide.

The % yield of this reaction is :-

(A) 59.5%

(B) 72.5%

(C) 97.5%

(D) 81.5%

Q19. MCQ (Single Correct Answer)

Open solution

The wavelength of light absorbed for the following complexes are in the order

$[Co(NH_3)_6]^{3+}$; $[Co(H_2O)_6]^{3+}$; $[Co(CN)_6]^{3-}$; $[Co(NH_3)_5(H_2O)]^{3+}$; $[CoF_6]^{3-}$

(I) (II) (III) (IV) (V)

(A) III $<$ IV $<$ I $<$ II $<$ V

(B) III $<$ I $<$ IV $<$ V $<$ II

(C) III $<$ I $<$ IV $<$ II $<$ V

(D) III $<$ I $<$ II $<$ IV $<$ V

Q20. MCQ (Single Correct Answer)

Open solution

Find out the statements which are not true.

A. Resonating structures with more number of covalent bonds and lesser charge separation are more stable.

B. In electromeric effect, an unsaturated system shows +E effect with nucleophile and -E effect with electrophile.

C. Inductive effect is responsible for high melting point, boiling point and dipole moment of polar compounds.

D. The greater the number of alkyl groups attached to the doubly bonded carbon atoms, higher is the heat of hydrogenation.

E. Stability of carbanion increases with the increase in $s-$ character of the carbon carrying the negative charge.

Choose the correct answer from the options given below :

(A) A, D & E only

(B) A, C & D only

(C) B & D only

(D) B, D & E only

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Q21. Numerical

Grignard reagent $\mathrm{RMgBr(P)}$ reacts with water and forms a gas $\mathrm{(Q)}$. One gram of Q occupies $1.4\mathrm{dm}^3$ at STP. (P) on reaction with dry ice in dry ether followed by $\mathrm{H_3O^+}$forms a compound (Z). 0.1 mole of (Z) will weigh ____ g. (Nearest integer)

Open solution

Q22. Numerical

Molar conductivity of a weak acid HQ of concentration 0.18 M was found to be $1/30$ of the molar conductivity of another weak acid HZ with concentration of 0.02 M . If $\lambda^\circ\mathrm{Q}^-$ happened to be equal with $\lambda^\circ\mathrm{Z}^-$, then the difference of the $\mathrm{pK_a}$ values of the two weak acids $(\mathrm{pK_a(HQ)} - \mathrm{pK_a(HZ)})$ is ____ (Nearest integer).

[Given: degree of dissociation $(\alpha) \ll 1$ for both weak acids, λ° : limiting molar conductivity of ions]

Open solution

Q23. Numerical

A chromium complex with a formula $\mathrm{CrCl_3 \cdot 6H_2O}$ has a spin only magnetic moment value of 3.87 BM and its solution conductivity corresponds to $1:2$ electrolyte. 2.75 g of the complex solution was initially passed through a cation exchanger. The solution obtained after the process was reacted with excess of $\mathrm{AgNO_3}$. The amount of AgCl formed in the above process is ____ g. (Nearest integer)

[Given: Molar mass in $\mathrm{gmol^{-1}Cr: 52; Cl: 35.5, Ag: 108, O: 16, H: 1}$]

Open solution

Q24. Numerical

The half-life of $^{65}\mathrm{Zn}$ is 245 days. After x days, 75% of original activity remained. The value of x in days is ____ . (Nearest integer)

(Given: $\log 3 = 0.4771$ and $\log 2 = 0.3010$)

Open solution

Q25. Numerical

0.25 g of an organic compound "A" containing carbon, hydrogen and oxygen was analysed using the combustion method. There was an increase in mass of $\mathrm{CaCl_2}$ tube and potash tube at the end of the experiment. The amount was found to be 0.15 g and 0.1837 g , respectively. The percentage of oxygen in compound A is

____ %. (Nearest integer)

(Given: molar mass in $\mathrm{gmol^{-1}H: 1, C: 12, O: 16}$)

Open solution

Mathematics 25 questions

Q26. MCQ (Single Correct Answer)

The smallest positive integral value of a, for which all the roots of $x^4 - ax^2 + 9 = 0$ are real and distinct, is equal to

(A) 7 (B) 3 (C) 4 (D) 9

Open solution

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Q27. MCQ (Single Correct Answer)

The letters of the word "UDAYPUR" are written in all possible ways with or without meaning and these words are arranged as in a dictionary. The rank of the word "UDAYPUR" is

(A) 1579 (B) 1578 (C) 1580 (D) 1581

Open solution

Q28. MCQ (Single Correct Answer)

Let a_1, a_2, a_3, a_4 be an A.P. of four terms such that each term of the A.P. and its common difference l are integers. If $a_1 + a_2 + a_3 + a_4 = 48$ and $a_1a_2a_3a_4 + l^4 = 361$, then the largest term of the A.P. is equal to

(A) 27 (B) 24 (C) 23 (D) 21

Open solution

Q29. MCQ (Single Correct Answer)

Consider the following three statements for the function $f : (0, \infty) \to \mathbb{R}$ defined by $f(x) = |\log_e x| - |x - 1|$:

(I) f is differentiable at all $x > 0$.

(II) f is increasing in $(0, 1)$.

(III) f is decreasing in $(1, \infty)$.

Then.

(A) Only (I) is TRUE. (B) Only (I) and (III) are TRUE.

(C) Only (II) and (III) are TRUE. (D) All (I), (II) and (III) are TRUE.

Open solution

Q30. MCQ (Single Correct Answer)

Let $P = [p_{ij}]$ and $Q = [q_{ij}]$ be two square matrices of order 3 such that $q_{ij} = 2^{(i+j-1)}p_{ij}$ and $\det(Q) = 2^{10}$. Then the value of $\det(\text{adj}(\text{adj } P))$ is:

(A) 81 (B) 16 (C) 124 (D) 32

Open solution

Q31. MCQ (Single Correct Answer)

Let $y = y(x)$ be a differentiable function in the interval $(0, \infty)$ such that $y(1) = 2$, and $\lim\limits_{t \to x} \left(\frac{t^2 y(x) - x^2 y(t)}{x - t} \right) = 3$ for each $x > 0$. Then $2y(2)$ is equal to :

(A) 27 (B) 18 (C) 23 (D) 12

Open solution

Q32. MCQ (Single Correct Answer)

The sum of all values of α, for which the shortest distance between the lines $\frac{x+1}{\alpha} = \frac{y-2}{-1} = \frac{z-4}{-\alpha}$ and $\frac{x}{\alpha} = \frac{y-1}{2} = \frac{z-1}{2\alpha}$ is $\sqrt{2}$, is

(A) -6 (B) -8 (C) 8 (D) 6

Open solution

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Q33. MCQ (Single Correct Answer)

Let $[t]$ denote the greatest integer less than or equal to t. If the function

$$f(x)=\begin{cases} b^2\sin\left(\frac{\pi}{2}\left[\frac{\pi}{2}(\cos x+\sin x)\cos x\right]\right), & x<0 \\ \frac{\sin x-\frac{1}{2}\sin 2x}{x^3} & ,x>0 \\ a & ,x=0 \end{cases}$$

is continuous at $x=0$, then a^2+b^2 is equal to :

(A) $\frac{1}{2}$ (B) $\frac{5}{8}$ (C) $\frac{3}{4}$ (D) $\frac{9}{16}$

Open solution

Q34. MCQ (Single Correct Answer)

Let $\mathrm{X}=\{x\in\ \mathrm{N}:1\le x\le 19\}$ and for some $a,b\in\mathbb{R}$, $\mathrm{Y}=\{ax+b:x\in\mathrm{X}\}$. If the mean and variance of the elements of Y are 30 and 750 , respectively, then the sum of all possible values of b is

(A) 20 (B) 100 (C) 80 (D) 60

Open solution

Q35. MCQ (Single Correct Answer)

The largest value of n, for which 40^n divides $60!$, is

(A) 14 (B) 13 (C) 11 (D) 12

Open solution

Q36. MCQ (Single Correct Answer)

Let $f(\alpha)$ denote the area of the region in the first quadrant bounded by $x=0, x=1, y^2=x$ and $y=|\alpha x-5|-|1-\alpha x|+\alpha x^2$. Then $(f(0)+f(1))$ is equal to

(A) 12 (B) 14 (C) 9 (D) 7

Open solution

Q37. MCQ (Single Correct Answer)

Let the image of parabola $x^2=4y$, in the line $x-y=1$ be $(y+a)^2=b(x-c)$, $a,b,c\in\ \mathrm{N}$. Then $a+b+c$ is equal to

(A) 12 (B) 8 (C) 6 (D) 4

Open solution

Q38. MCQ (Single Correct Answer)

Let the angles made with the positive x-axis by two straight lines drawn from the point $\mathrm{P}(2,3)$ and meeting the line $x+y=6$ at a distance $\sqrt{\frac{2}{3}}$ from the point P be θ_1 and θ_2. Then the value of $(\theta_1+\theta_2)$ is:

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{12}$ (D) $\frac{\pi}{6}$

Open solution

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Q39. MCQ (Single Correct Answer)

If the domain of the function $f(x)=\sin^{-1}\left(\frac{1}{x^2-2x-2}\right)$, is $(-\infty,\alpha]\cup[\beta,\gamma]\cup[\delta,\infty)$, then $\alpha+\beta+\gamma+\delta$ is equal to

(A) 4 (B) 2 (C) 5 (D) 3

Open solution

Q40. MCQ (Single Correct Answer)

Let $f(x)=\int\frac{7x^{10}+9x^8}{(1+x^2+2x^9)^2}dx, x>0, \lim\limits_{x\to 0} f(x)=0$ and $f(1)=\frac{1}{4}$.

If $\mathrm{A}=\begin{bmatrix}0 & 0 & 1\\ \frac{1}{4} & f'(1) & 1\\ \alpha^2 & 4 & 1\end{bmatrix}$ and $\mathrm{B}=\mathrm{adj}(\mathrm{adj}\,\mathrm{A})$ be such that $|\mathrm{B}|=81$, then α^2 is equal to

(A) 2 (B) 4 (C) 3 (D) 1

Open solution

Q41. MCQ (Single Correct Answer)

Let the length of the latus rectum of an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1,(a>b)$, be 30 . If its eccentricity is the maximum value of the function $f(t)=-\frac{3}{4}+2t-t^2$, then $\left(a^2+b^2\right)$ is equal to

(A) 276 (B) 516 (C) 256 (D) 496

Open solution

Q42. MCQ (Single Correct Answer)

Let f be a function such that $3f(x)+2f\left(\frac{m}{19x}\right)=5x, x\neq 0$, where $m=\sum\limits_{i=1}^{9}(i)^2$. Then $f(5)-f(2)$ is equal to

(A) 36 (B) 9 (C) -9 (D) 18

Open solution

Q43. MCQ (Single Correct Answer)

Let $\vec{a}=2\hat{\mathrm{i}}-\hat{\mathrm{j}}-\hat{\mathrm{k}}, \vec{b}=\hat{\mathrm{i}}+3\hat{\mathrm{j}}-\hat{\mathrm{k}}$ and $\vec{c}=2\hat{\mathrm{i}}+\hat{\mathrm{j}}+3\hat{\mathrm{k}}$. Let $\vec{v}$ be the vector in the plane of the vectors $\vec{a}$ and $\vec{b}$, such that the length of its projection on the vector $\vec{c}$ is $\frac{1}{\sqrt{14}}$. Then $|\vec{v}|$ is equal to

(A) $\frac{\sqrt{35}}{2}$ (B) 13 (C) $\frac{\sqrt{21}}{2}$ (D) 7

Open solution

Q44. MCQ (Single Correct Answer)

Let $\vec{a}=2\hat{\mathrm{i}}-5\hat{\mathrm{j}}+5\hat{\mathrm{k}}$ and $\vec{b}=\hat{\mathrm{i}}-\hat{\mathrm{j}}+3\hat{\mathrm{k}}$. If $\vec{c}$ is a vector such that $2(\vec{a}\times\vec{c})+3(\vec{b}\times\vec{c})=\vec{0}$ and $(\vec{a}-\vec{b})\cdot\vec{c}=-97$, then $|\vec{c}\times\hat{\mathrm{k}}|^2$ is equal to

(A) 193 (B) 205 (C) 218 (D) 233

Open solution

Q45. MCQ (Single Correct Answer)

$\left(\frac{1}{3}+\frac{4}{7}\right)+\left(\frac{1}{3^2}+\frac{1}{3}\times\frac{4}{7}+\frac{4^2}{7^2}\right)+\left(\frac{1}{3^3}+\frac{1}{3^2}\times\frac{4}{7}+\frac{1}{3}\times\frac{4^2}{7^2}+\frac{4^3}{7^3}\right)+\ldots$ upto infinite terms, is equal to

(A) $\frac{7}{4}$ (B) $\frac{4}{3}$ (C) $\frac{6}{5}$ (D) $\frac{5}{2}$

Open solution

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Q46. Numerical

Let S be a set of 5 elements and $\mathrm{P}(\mathrm{S})$ denote the power set of S . Let E be an event of choosing an ordered pair (A, B) from the set $\mathrm{P}(\mathrm{S}) \times \mathrm{P}(\mathrm{S})$ such that $\mathrm{A} \cap \mathrm{B} = \emptyset$. If the probability of the event E is $\frac{3^p}{2^q}$, where $p, q \in N$, then $p + q$ is equal to

Open solution

Q47. Numerical

Let $z = (1+i)(1+2i)(1+3i)\ldots.(1+ni)$, where $i = \sqrt{-1}$. If $|z|^2 = 44200$, then n is equal to ____

Open solution

Q48. Numerical

The number of elements in the set $\{x \in [0, 180^\circ] : \tan(x + 100^\circ) = \tan(x + 50^\circ)\tan x \tan(x - 50^\circ)\}$ is ____ .

Open solution

Q49. Numerical

Let (h, k) lie on the circle $\mathrm{C} : x^2 + y^2 = 4$ and the point $(2h + 1, 3k + 2)$ lie on an ellipse with eccentricity e. Then the value of $\frac{5}{e^2}$ is equal to ____ .

Open solution

Q50. Numerical

If $f(x)$ satisfies the relation $f(x) = e^x + \int_0^1 (y + xe^x) f(y)dy$, then $e + f(0)$ is equal to ____ .

Open solution

Physics 25 questions

Q51. MCQ (Single Correct Answer)

The fifth harmonic of a closed organ pipe is found to be in unison with the first harmonic of an open pipe. The ratio of lengths of closed pipe to that of the open pipe is $5/x$. The value of x is ____

(A) 3 (B) 2 (C) 4 (D) 1

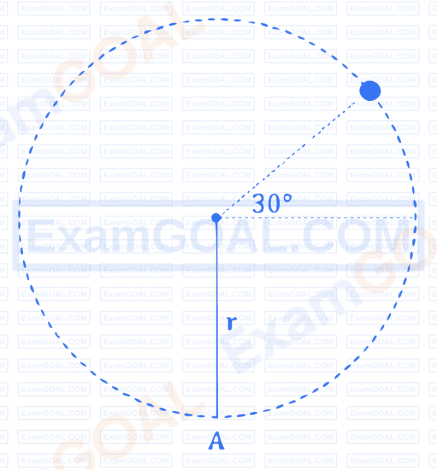
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Q52. MCQ (Single Correct Answer)

Open solution

In case of vertical circular motion of a particle by a thread of length r if the tension in the thread is zero at an angle 30° shown in figure, the velocity at the bottom point (A) of the circular path is (g = gravitational acceleration)



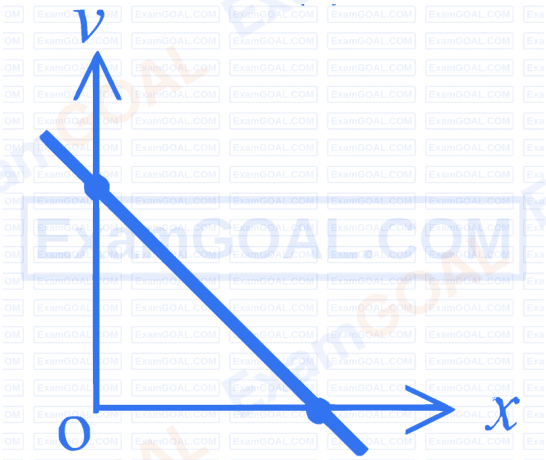
(A) $\sqrt{\frac{7}{2}gr}$ (B) $\sqrt{4gr}$ (C) $\sqrt{\frac{5}{2}gr}$ (D) $\sqrt{5gr}$

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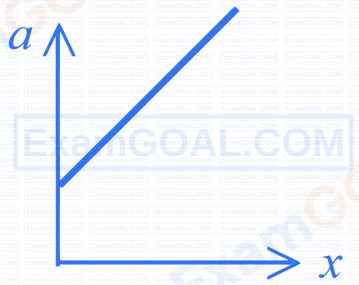
Q53. MCQ (Single Correct Answer)

Open solution

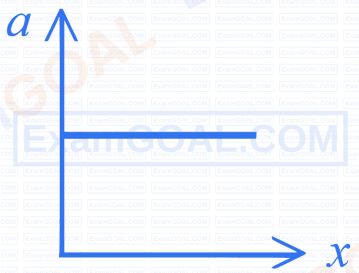
The velocity (v) - Distance (x) graph is shown in figure. Which graph represents acceleration(a) versus distance (x) variation of this system?



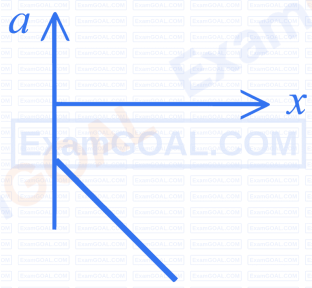
(A)



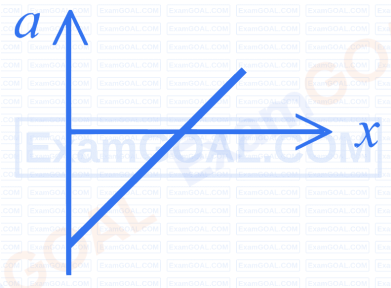
(B)



(C)



(D)



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Q54. MCQ (Single Correct Answer)

Open solution

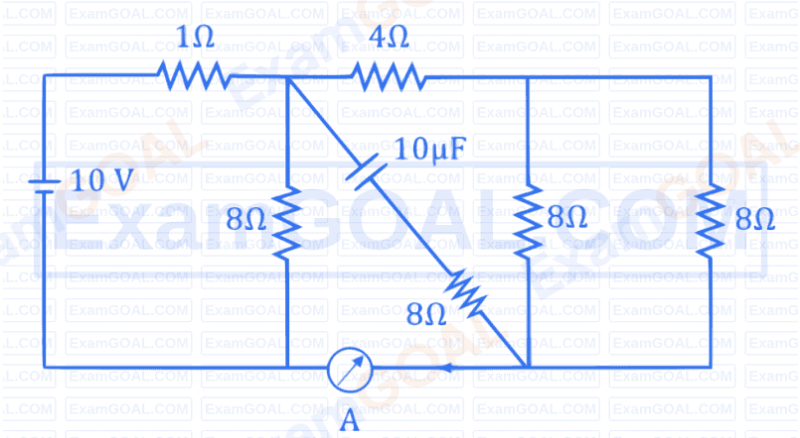
In the Young's double slit experiment the intensity produced by each one of the individual slits is I_o. The distance between two slits is 2 mm . The distance of screen from slits is 10 m . The wavelength of light is $6000\ \mathrm{A}^{\circ}$. The intensity of light on the screen in front of one of the slits is ____

(A) $\frac{I_o}{2}$ (B) I_o (C) $2I_o$ (D) $4I_o$

Q55. MCQ (Single Correct Answer)

Open solution

The reading of the ammeter (A) in steady state in the following circuit (assuming negligible internal resistance of the ammeter) is ____ A.



(A) 1/2 (B) 1 (C) 2 (D) 0

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Q56. MCQ (Single Correct Answer)

Open solution

Identify the correct truth table of the given logic circuit.



(A)

A	B	Y
0	0	0
0	1	0
1	0	1
1	1	0

(B)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

(C)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

(D)

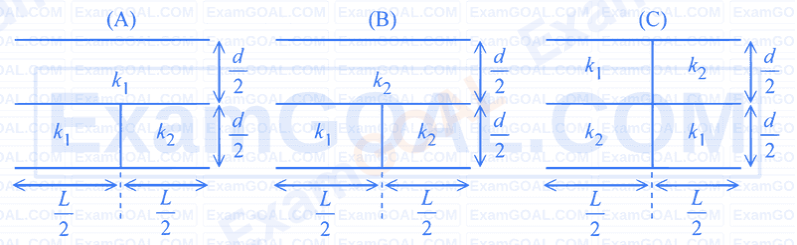
A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

Q57. MCQ (Single Correct Answer)

Open solution

Three parallel plate capacitors each with area A and separation d are filled with two dielectrics (k_1 and k_2) in the following fashion. Which of the following is true?

$(k_1 > k_2)$



(A) $C_C > C_B > C_A$ **(B)** $C_B > C_C > C_A$ **(C)** $C_A > C_C > C_B$ **(D)** $C_C > C_A > C_B$

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Q58. MCQ (Single Correct Answer)

A moving coil galvanometer of resistance 100Ω shows a full scale deflection for a current of 1 mA . The value of resistance required to convert this galvanometer into an ammeter, showing full scale deflection for a current of 5 mA , is ____ Ω

Open solution

(A) 25 (B) 2.5 (C) 0.5 (D) 10

Q59. MCQ (Single Correct Answer)

Distance between an object and three times magnified real image is 40 cm . The focal length of the mirror used is ____ cm .

Open solution

(A) -20 (B) $-15/2$ (C) -10 (D) -15

Q60. MCQ (Single Correct Answer)

A cubical block of density $\rho_b = 600\ \mathrm{kg/m^3}$ floats in a liquid of density $\rho_e = 900\mathrm{kg/m^3}$. If the height of block is $H = 8.0\ \mathrm{cm}$ then height of the submerged part is ____ cm .

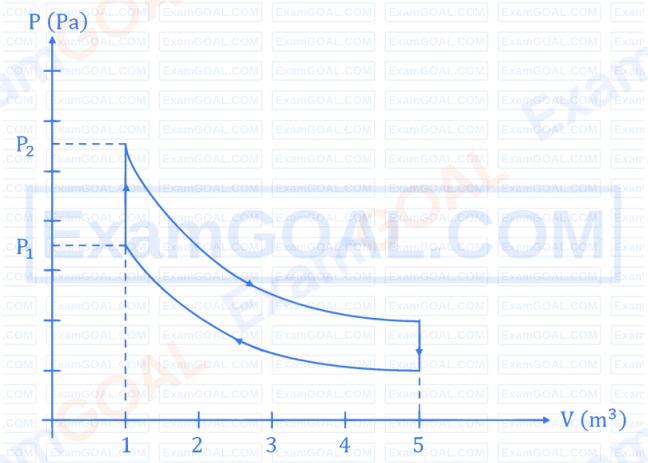
Open solution

(A) 6.3 (B) 4.3 (C) 7.3 (D) 5.3

Q61. MCQ (Single Correct Answer)

10 mole of an ideal gas is undergoing the process shown in the figure. The heat involved in the process from P_1 to P_2 is α Joule ($P_1 = 21.7\ \mathrm{Pa}$ and $P_2 = 30\ \mathrm{Pa}, \mathrm{C}_v = 21\ \mathrm{J/K.mol}, R = 8.3\ \mathrm{J/mol.K}$). The value of α is ____ .

Open solution



(A) 21 (B) 28 (C) 24 (D) 15

Q62. MCQ (Single Correct Answer)

In a vernier callipers, 50 vernier scale divisions are equal to 48 main scale divisions. If one main scale division $= 0.05\ \mathrm{mm}$, then the least count of the vernier callipers is ____ mm.

Open solution

(A) 0.002 (B) 0.02 (C) 0.05 (D) 0.005

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Q63. MCQ (Single Correct Answer)

Open solution

Five persons P_1, P_2, P_3, P_4 and P_5 recorded object distance (u) and image distance (v) using same convex lens having power +5 D as $(25, 96), (30, 62), (35, 37), (45, 35)$ and $(50, 32)$ respectively. Identify correct statement

(A) Readings recorded by P_3 and P_2 persons are incorrect

(B) Readings recorded by all persons are correct

(C) Readings recorded by P_3 person are incorrect

(D) Readings recorded by P_4 and P_5 persons are incorrect

Q64. MCQ (Single Correct Answer)

Open solution

The binding energy for the following nuclear reactions are expressed in MeV .

$_2\mathrm{He}^3 + {}_0\mathrm{n}^1 \rightarrow {}_2\mathrm{He}^4 + 20\mathrm{MeV}$

$_2\mathrm{He}^4 + {}_0\mathrm{n}^1 \rightarrow {}_2\mathrm{He}^5 - 0.9\mathrm{MeV}$

If $\mathrm{X}_3, \mathrm{X}_4, \mathrm{X}_5$ denote the stability of $_2\mathrm{He}^3$, $_2\mathrm{He}^4$ and $_2\mathrm{He}^5$, respectively, then the correct order is :

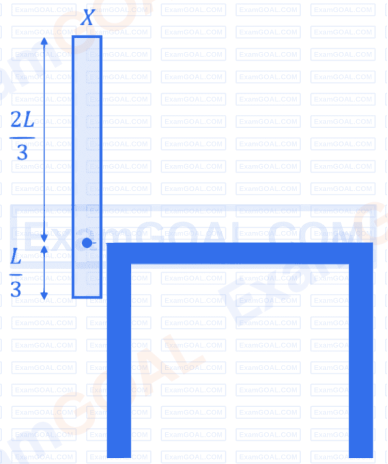
(A) $X_4 > X_5 > X_3$ (B) $X_4 < X_5 < X_3$ (C) $X_4 > X_5 < X_3$ (D) $X_4 = X_5 = X_3$

Q65. MCQ (Single Correct Answer)

Open solution

A thin uniform rod (X) of mass M and length L is pivoted at a height $\left(\frac{L}{3}\right)$ as shown in the figure. The rod is allowed to fall from a vertical position and lie horizontally on the table. The angular velocity of this rod when it hits the table top, is ____ .

(g = gravitational acceleration)



(A) $\sqrt{\frac{3}{2}\frac{g}{L}}$ (B) $\sqrt{\frac{3g}{L}}$

(C) $\frac{3}{\sqrt{2}}\sqrt{\frac{g}{L}}$ (D) $\frac{1}{\sqrt{2}}\sqrt{\frac{g}{L}}$

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Q66. MCQ (Single Correct Answer)

A point source is kept at the center of a spherically enclosed detector. If the volume of the detector increased by 8 times, the intensity will

(A) increase by 64 times
(B) decrease by 4 times
(C) decrease by 8 times
(D) increase by 8 times

Open solution

Q67. MCQ (Single Correct Answer)

A regular hexagon is formed by six wires each of resistance $r\Omega$ and the corners are joined to the centre by wires of same resistance. If the current enters at one corner and leaves at the opposite corner, the equivalent resistance of the hexagon between the two opposite corners will be

(A) $\frac{3}{5}r$
(B) $\frac{4}{5}r$
(C) $\frac{5}{8}r$
(D) $\frac{3}{4}r$

Open solution

Q68. MCQ (Single Correct Answer)

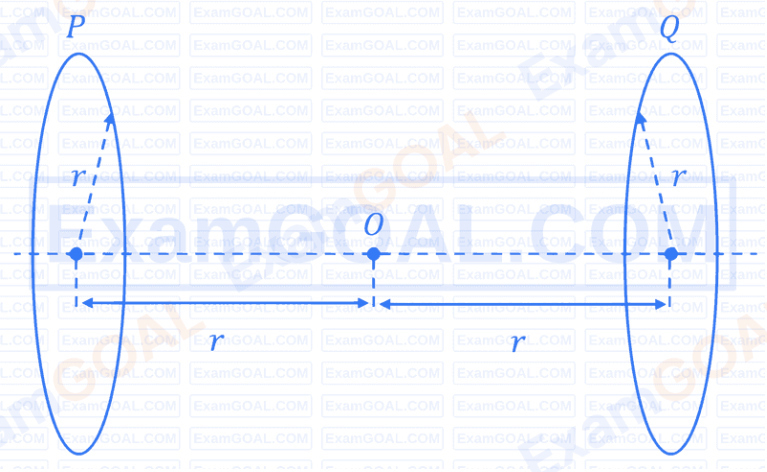
A flexible chain of mass m hangs between two fixed points at the same level. The inclination of the chain with the horizontal at the two points of support is 30°. Considering the equilibrium of each half of the chain, the tension of the chain at the lowest point is ____ .

(A) $m\,\mathrm{g}$
(B) $\frac{\sqrt{3}}{2}mg$
(C) $\frac{1}{2}mg$
(D) $\sqrt{3}m\,\mathrm{g}$

Open solution

Q69. MCQ (Single Correct Answer)

Two identical circular loops P and Q each of radius r are lying in parallel planes such that they have common axis. The current through P and Q are I and $4I$ respectively in clockwise direction as seen from O. The net magnetic field at O is :



(A) $\frac{\mu_o I}{4\sqrt{2}r}$ towards Q
(B) $\frac{3\mu_o I}{4\sqrt{2}r}$ towards Q
(C) $\frac{3\mu_0 I}{4\sqrt{2}r}$ towards P
(D) $\frac{\mu_o I}{4\sqrt{2}r}$ towards P

Open solution

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Q70. MCQ (Single Correct Answer)

When a light of a given wavelength falls on a metallic surface the stopping potential for photoelectrons is 3.2 V . If a second light having wavelength twice of first light is used, the stopping potential drops to 0.7 V . The wavelength of first light is ____ m .

$\left(\mathrm{h}=6.63 \times 10^{-34}\ \mathrm{J.s}, \mathrm{e}=1.6 \times 10^{-19} \mathrm{C}, \mathrm{c}=3 \times 10^{8}\ \mathrm{m/s}\right)$

(A) 2.9×10^{-8} (B) 2.5×10^{-7} (C) 3.1×10^{-7} (D) 2.2×10^{-8}

Open solution

Q71. Numerical

In a meter bridge experiment to determine the value of unknown resistance, first the resistances 2Ω and 3Ω are connected in the left and right gaps of the bridge and the null point is obtained at a distance l cm from the left. Now when an unknown resistance $x\Omega$ is connected in parallel to 3Ω resistance, the null point is shifted by 10 cm to the right of wire. The value of unknown resistance x is

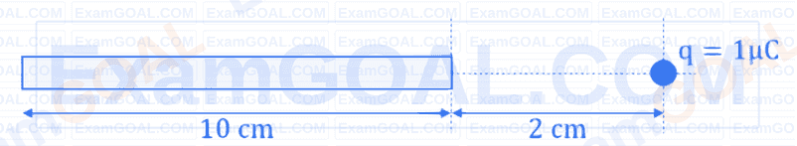
____ Ω.

Open solution

Q72. Numerical

A point charge $q = 1\mu\mathrm{C}$ is located at a distance 2 cm from one end of a thin insulating wire of length 10 cm having a charge $Q = 24\mu\mathrm{C}$, distributed uniformly along its length, as shown in figure. Force between q and wire is ____ N.

(Use : $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9\ \mathrm{N \cdot m^2/C^2}$)



Open solution

Q73. Numerical

A soap bubble of surface tension $0.04\ \mathrm{N/m}$ is blown to a diameter of 7 cm . If $(15000 - x)\mu\mathrm{J}$ of work is done in blowing it further to make its diameter 14 cm , then the value of x is ____ .

$(\pi = 22/7)$

Open solution

Q74. Numerical

When 300 J of heat given to an ideal gas with $C_p = \frac{7}{2}R$ its temperature raises from $20^\circ\mathrm{C}$ to $50^\circ\mathrm{C}$ keeping its volume constant. The mass of the gas is (approximately) ____ g. $(\mathrm{R} = 8.314\ \mathrm{J/mol.\ K})$

Open solution

Q75. Numerical

A uniform solid cylinder of length L and radius R has moment of inertia about its axis equal to I_1. A small co-centric cylinder of length $L/2$ and radius $R/3$ carved from this cylinder has moment of inertia about its axis equals to I_2. The ratio I_1/I_2 is ____ .

Open solution

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Answer Key

Chemistry

1. **D**	2. **D**	3. **D**	4. **A**	5. **D**
6. **D**	7. **B**	8. **B**	9. **A**	10. **B**
11. **C**	12. **A**	13. **A**	14. **A**	15. **B**
16. **D**	17. **C**	18. **D**	19. **C**	20. **C**
21. **6**	22. **2**	23. **3**	24. **102**	25. **73**

Mathematics

26. **A**	27. **C**	28. **A**	29. **B**	30. **B**
31. **C**	32. **A**	33. **C**	34. **D**	35. **A**
36. **D**	37. **C**	38. **A**	39. **A**	40. **B**
41. **D**	42. **D**	43. *Scan QR*	44. **C**	45. **D**
46. **15**	47. **5**	48. **4**	49. **9**	50. **2**

Physics

51. **B**	52. **A**	53. **D**	54. **B**	55. **B**
56. **A**	57. **C**	58. **A**	59. **D**	60. **D**
61. **A**	62. **A**	63. **C**	64. **A**	65. **B**
66. **B**	67. **B**	68. **B**	69. **B**	70. **B**
71. **6**	72. **90**	73. **11304**	74. **481**	75. **162**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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